

Finding Zeros of Quadratic Functions Algebraically

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A **zero** of a quadratic function is the value of x that makes the function equal to zero.

Factored Form

A quadratic function in **factored form** is written as

$$f(x) = a(x - r_1)(x - r_2)$$

where the zeros are r_1 and r_2 .

Zero Product Property

The **Zero Product Property** states:

$$\text{if } ab = 0, \text{ then } a = 0 \text{ or } b = 0$$

This lets us find zeros directly from factored form.

Standard Form

A quadratic function in **standard form** is written as

$$f(x) = ax^2 + bx + c$$

Solving by Factoring

To find the zeros of a quadratic function in standard form, first solve the equation

$$ax^2 + bx + c = 0$$

by **factoring** whenever possible.

Zeros and x -Intercepts

The zeros correspond to the x -**intercepts** of the graph of the quadratic function.

Types of Zeros

A quadratic function may have:

- ▶ two distinct zeros,
- ▶ one repeated zero, or
- ▶ no real zero.

Examples

Example 1: Factored Form

Find the zeros of

$$f(x) = (x - 3)(x + 2)$$

By the Zero Product Property:

$$x - 3 = 0 \quad \text{or} \quad x + 2 = 0$$

$$x = 3 \quad \text{or} \quad x = -2$$

Example 2: Standard Form (Two Zeros)

Find the zeros of

$$f(x) = x^2 - 5x + 6$$

Factor:

$$f(x) = (x - 2)(x - 3)$$

$$x = 2 \quad \text{or} \quad x = 3$$

Example 3: Repeated Zero

Find the zero(s) of

$$f(x) = x^2 - 6x + 9$$

Factor:

$$f(x) = (x - 3)^2$$

$$x = 3$$

This is a **repeated zero**.

Practice Exercises

Practice 1

Find the zeros.

$$f(x) = (x - 5)(x + 2)$$

Practice 2

Find the zeros.

$$f(x) = 2(x + 4)(x - 1)$$

Practice 3

Find the zeros.

$$f(x) = -(x - 6)(x - 3)$$

Practice 4

Find the zeros.

$$f(x) = 3(x + 1)^2$$

Practice 5

Find the zeros.

$$f(x) = 5(x - 7)^2$$

Practice 6

Find the zeros.

$$f(x) = x^2 - 9x + 20$$

Practice 7

Find the zeros.

$$f(x) = 2x^2 - 7x + 3$$

Practice 8

Find the zeros.

$$f(x) = x^2 + 5x + 6$$

Practice 9

Find the zeros.

$$f(x) = 3x^2 - 12x + 9$$

Practice 10

Find the zeros.

$$f(x) = 4x^2 + 4x - 15$$

Activities

Activity 1

Find the zeros.

$$f(x) = (x - 4)(x + 3)$$

Activity 2

Find the zeros.

$$f(x) = 5(x - 2)(x + 1)$$

Activity 3

Find the zeros.

$$f(x) = -2(x - 8)(x - 5)$$

Activity 4

Find the zeros.

$$f(x) = 4(x - 3)^2$$

Activity 5

Find the zeros.

$$f(x) = 2(x + 6)^2$$

Activity 6

Find the zeros.

$$f(x) = x^2 - 11x + 28$$

Activity 7

Find the zeros.

$$f(x) = 2x^2 - 5x - 3$$

Activity 8

Find the zeros.

$$f(x) = x^2 + 7x + 12$$

Activity 9

Find the zeros.

$$f(x) = 2x^2 - 8x + 8$$

Activity 10

Find the zeros.

$$f(x) = 3x^2 + 2x - 8$$

Thank you!